

Problem Set 3

Applied Stats/Quant Methods 1

Due: November 13, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in **R**, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 13, 2025. No late assignments will be accepted.

In this problem set, you will run several regressions and create an add variable plot (see the lecture slides) in **R** using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```

1 vote_share_model <-lm(inc.sub$voteshare~inc.sub$difflog)
2 summary(vote_share_model)

```

```

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.579031    0.002251  257.19  <2e-16 ***
inc.sub$difflog 0.041666    0.000968   43.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic: 1853 on 1 and 3191 DF,  p-value: < 2.2e-16

```

2. Make a scatterplot of the two variables and add the regression line.

```

1 pdf("./voteshare_by_difflog.pdf")
2 plot(inc.sub$difflog, inc.sub$voteshare,
3      main="Difference in Campaign Spending vs Incumbant Vote Share",
4      xlab="Difference in Spending",
5      ylab="Incumbant Vote Share")
6 abline(lm(inc.sub$voteshare~inc.sub$difflog), col='blue')
7
8 dev.off()

```

See Figure 1.

3. Save the residuals of the model in a separate object.

```

1 vote_share_model_residuals <- vote_share_model$residuals

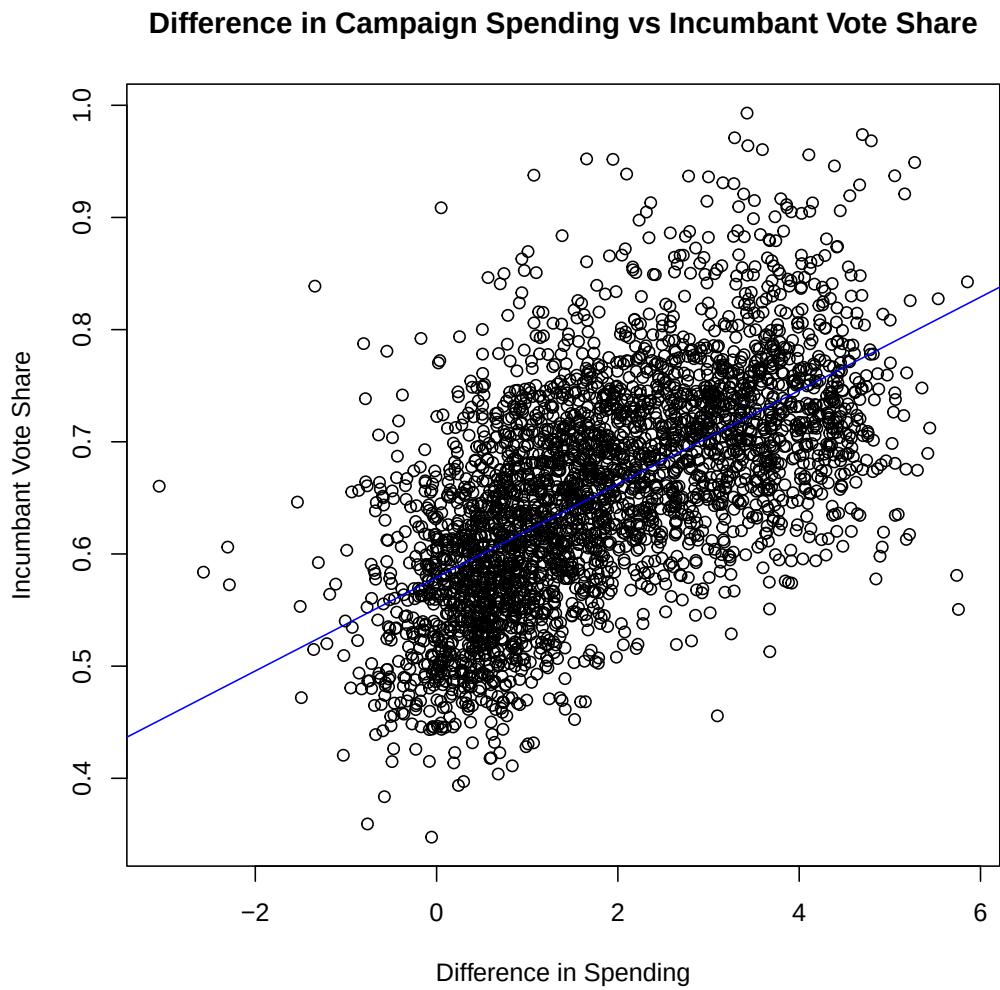
```

4. Write the prediction equation.

$$Y = 0.579 + 0.041X + 0.079$$

]

Figure 1: Difference in Campaign Spending vs Incumband Vote Share.



Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is **presvote** and the explanatory variable is **difflog**.

```
1 pres_vote_model <- lm(inc.sub$presvote ~ inc.sub$difflog)
2 summary(pres_vote_model)
```

```

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.507583   0.003161  160.60   <2e-16 ***
inc.sub$difflog 0.023837   0.001359   17.54   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1104 on 3191 degrees of freedom
Multiple R-squared:  0.08795, Adjusted R-squared:  0.08767
F-statistic: 307.7 on 1 and 3191 DF,  p-value: < 2.2e-16

```

2. Make a scatterplot of the two variables and add the regression line.

```

1 pdf("./presvote_by_difflog.pdf")
2 plot(inc.sub$difflog, inc.sub$presvote,
3      main="Difference in Campaign Spending vs Presidential Vote ",
4      xlab="Difference in Spending",
5      ylab="Presidential Vote")
6 abline(lm(inc.sub$presvote ~ inc.sub$difflog), col='blue')
7 dev.off()

```

See Figure 2.

3. Save the residuals of the model in a separate object.

```

1 pres_vote_model_residuals <- pres_vote_model$residuals

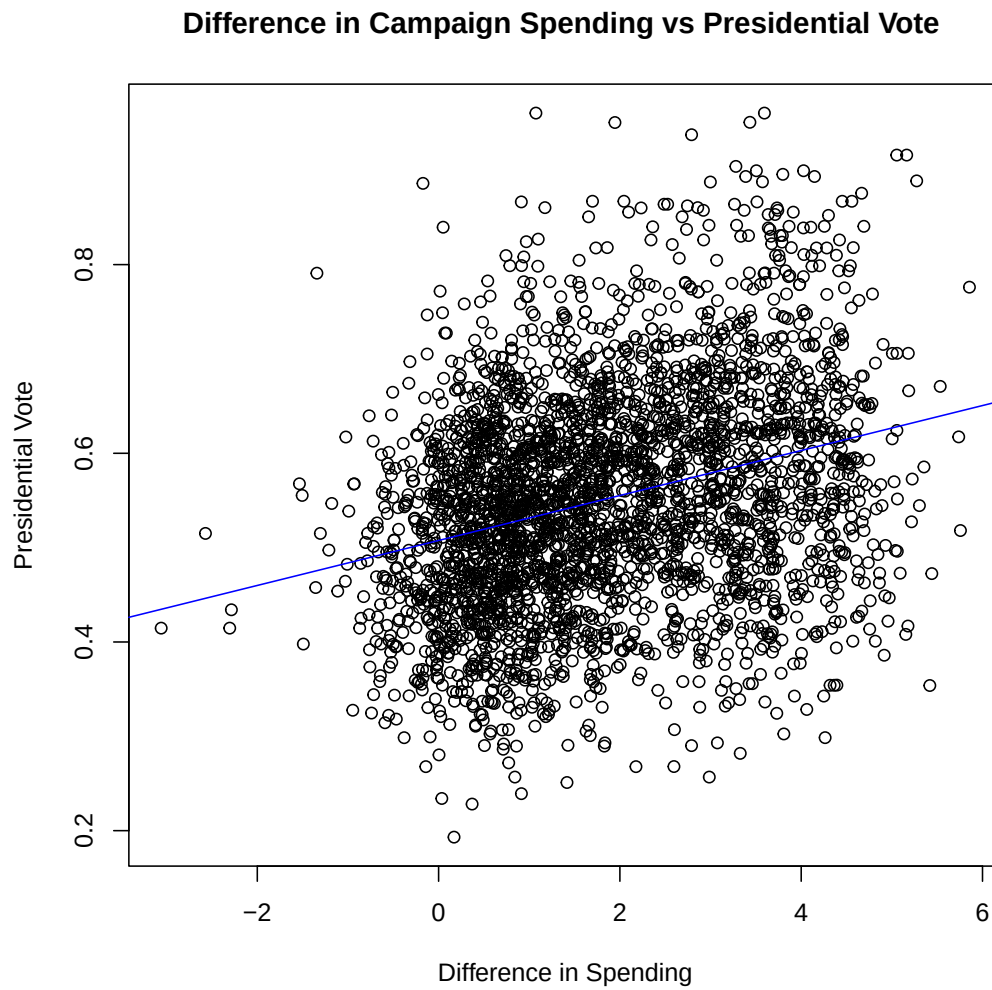
```

4. Write the prediction equation.

$$Y = 0.507 + 0.024X + 0.110$$

]

Figure 2: Difference in Campaign Spending vs Presidential Vote.



Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```
1 vote_share_presvote_model <- lm(inc.sub$voteshare ~ inc.sub$presvote)
2 summary(vote_share_model)
```

Coefficients:

```

Estimate Std. Error t value Pr(>|t|)
(Intercept)    0.579031    0.002251  257.19  <2e-16 ***
inc.sub$difflog 0.041666    0.000968   43.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07867 on 3191 degrees of freedom
Multiple R-squared:  0.3673, Adjusted R-squared:  0.3671
F-statistic: 1853 on 1 and 3191 DF,  p-value: < 2.2e-16

```

2. Make a scatterplot of the two variables and add the regression line.

```

1 pdf("./voteshare_by_presvote.pdf")
2 plot(inc.sub$presvote, inc.sub$voteshare,
3       main="Presidential Vote vs Incumbant Vote Share",
4       xlab="Presidential Vote",
5       ylab="Incumbant Vote Share")
6 abline(lm(inc.sub$voteshare~inc.sub$presvote), col='blue')
7
8 dev.off()

```

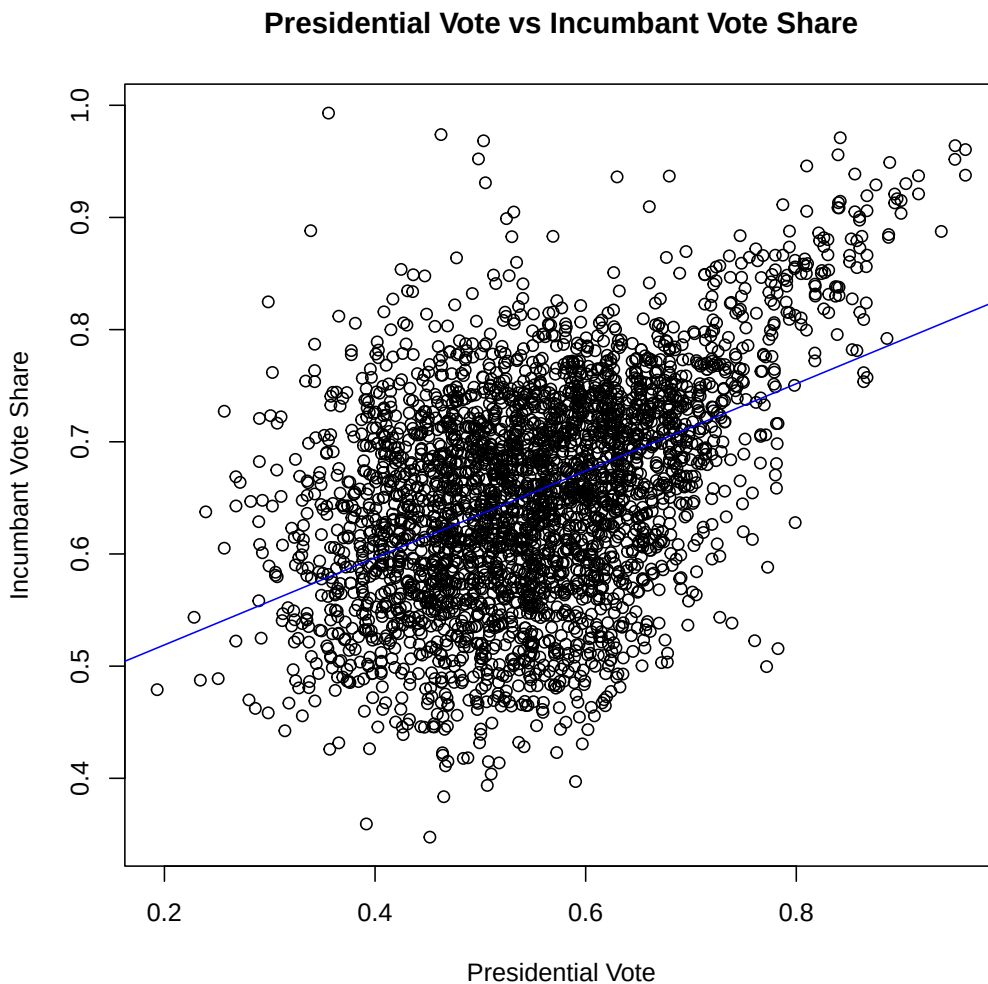
See Figure 3.

3. Write the prediction equation.

$$Y = 0.579 + 0.042X + 0.078$$

]

Figure 3: Presidential Vote vs Incumbant Vote Share.



Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```
1 residuals_regression <- lm(vote_share_model_residuals ~ pres_vote_model_residuals)
```

```
2 summary(residuals_regression)
```

```
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      -2.177e-18  1.299e-03    0.00      1
pres_vote_model_residuals  2.569e-01  1.176e-02   21.84 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07338 on 3191 degrees of freedom
Multiple R-squared:  0.13, Adjusted R-squared:  0.1298
F-statistic:  477 on 1 and 3191 DF,  p-value: < 2.2e-16
```

2. Make a scatterplot of the two residuals and add the regression line.

```
1 pdf("./presvote_residuals_by_vote_share_residuals.pdf")
2 plot(pres_vote_model_residuals, vote_share_model_residuals,
3      main="PresVote residuals vs Incumbant Vote Share residuals",
4      xlab="PresVote residuals",
5      ylab="Incumbant Vote Share residuals")
6 abline(lm(vote_share_model_residuals~pres_vote_model_residuals), col='blue'
7         ')
8 dev.off()
```

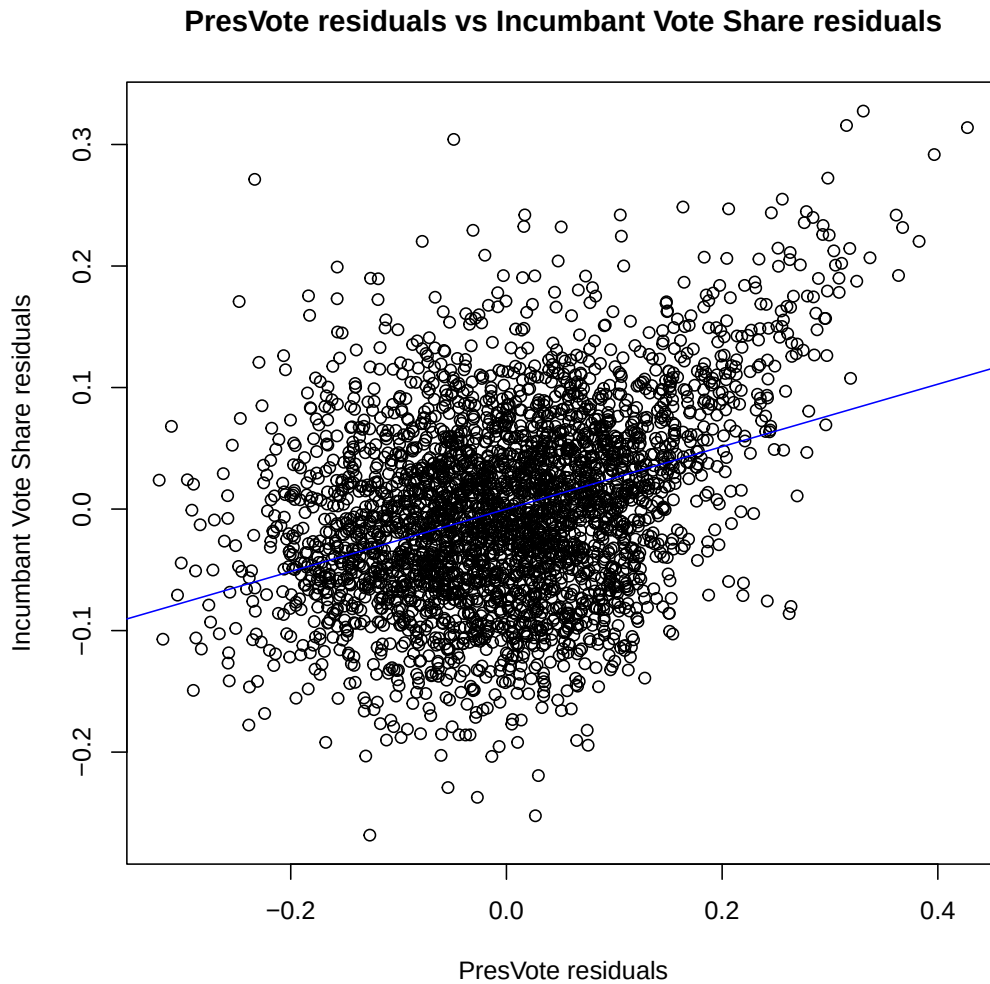
See Figure 4.

3. Write the prediction equation.

$$Y = 2.177 \times 10^{-18} + 0.257X + 0.073$$

]

Figure 4: PresVote residuals vs Incumbant Vote Share residuals.



Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```
1 residuals_regression <- lm(vote_share_model_residuais ~ pres_vote_model_
  residuals)
2 summary(residuals_regression)
```

```

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept)      0.4486442  0.0063297   70.88  <2e-16 ***
inc.sub$difflog   0.0355431  0.0009455   37.59  <2e-16 ***
inc.sub$presvote  0.2568770  0.0117637   21.84  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.07339 on 3190 degrees of freedom
Multiple R-squared:  0.4496, Adjusted R-squared:  0.4493
F-statistic: 1303 on 2 and 3190 DF,  p-value: < 2.2e-16

```

2. Write the prediction equation.

$$Y = 0.449 + 0.355X_1 + 0.257X_2 + 0.073$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

In questions 4 and 5, the standard error terms, ϵ , are equal. This is a strong indicator that X_1 , difflog , and X_1 , presvote , are uncorrelated. Since:

"The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

The residuals from the regression of $Q1_residuals \sim Q2_residuals$ tells us

How much of the variation in Y that X_1 didn't explain can be 'predicted' by the variation in Y that X_2 didn't explain?"

in this case, since the standard error terms are the same, we can expect that X_1 and X_2 are uncorrelated.